An Analysis of Spatial Clustering in MLB Batted Ball Distributions

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Introduction

In the spring of 2023, Major League Baseball implemented one of the most consequential rule changes in the modern era: a ban on the defensive shift. Before then, teams had increasingly relied on radical alignments such as stacking three infielders on one side of second base ti neutralize pull hitters. By restricting defenses to two infielders on each side of second base, the question of where hitters actually hit the ball becomes more important. The spatial distribution of batted balls is once again the primary contest between hitters and fielders.

Understanding how Major League Baseball (MLB) players hits are distributed across the diamond is central to evaluating defensive strategies. While traditional statistics summarize performance outcomes, they often overlook the spatial dimension of hitting. Without formal spatial techniques, it remains unclear as to whether hitters exhibit genuine clustering in their batted ball locations.

This project investigates whether MLB hitters exhibit spatial clustering in their batted ball locations, and how these patterns of concentration vary among a selection of high-volume hitters. We are using data from the 2023 MLB regular season. High-volume hitters will be defined as any player with a minimum of five-hundred plate appearances and a minimum of one-hundred batted balls in play.

To implement this analysis, we will employ spatial clustering methods such as density-based clustering and kernel density estimation to identify regions of high batted ball concentration. Then we will assess statistical significance through a series of permutations, determining whether clustering patterns deviate from randomness.

This project's primary data source is Baseball Savant, the official repository for MLB Statcast data. This source was chosen because it is the only source of pitch-level tracked batted-ball location data for MLB. It includes batted ball coordinates, player names, game dates, batter handedness, and event-level descriptive statistics. FanGraphs directional hitting data were also used to classify hitter tendencies as pull, center, opposite-field, or balanced. By focusing this analysis on the 2023 MLB regular season, the defensive shift restrictions create a more standardized environment for studying batted-ball distributions.

This project's study area is the fair-territory region of the baseball field, including home runs. While this is not a traditional study area, it functions as a bounded spatial region with meaningful subregions, including the infield, left field, center field, and right field. The baseball field overlaid on the visualizations uses an approximation for its dimensions because MLB ballparks vary in outfield wall distances and boundary shapes.

By treating the baseball field as a continuous spatial domain, this analysis offers a more rigorous foundation for the kinds of positioning decisions that teams face.

Literature Background

The increasing availability of high-resolution tracking data has transformed the metrics and analysis of performance in Major League Baseball. Statcast, as described by the MLB is a "state-of-the-art tracking technology" that enables researchers to investigate aspects of gameplay beyond the traditional box scores "in ways that were never possible in the past" (MLB 2026). Some of these granular aspects include swing mechanics, and ball flight. Statcast was originally implemented in 2015 consisting of a combination of cameras and radar systems, however in 2020, the MLB added the Hawk-Eye, a high-speed camera allowing for a replay system and various new features.

One substantial body of work examines batting performance and offensive prediction, leveraging Statcast data including "launch angle, launch velocity, and hit distance" (Bailey 2017). Using logistic regression to estimate the hit probabilities, these predictions are then aggregated to forecast future batting averages. Importantly, Bailey's analysis utilized Statcast data in combination with Player Empirical Comparison and Optimization Test Algorithm (PECOTA), a prediction system that considers previous seasons to predict batting averages. A linear regression was conducted on the relationship between the previous PECOTA and new Statcast data to create an even stronger metric for predicting batting averages.

Gergis (2024) provides a recent and comprehensive analysis of on-base percentage, with a machine learning approach and 2015-2023 Statcast data. The analysis leverages over one million pitch observations, and compares traditional statistical methods such as logistic regression, to the more advanced techniques of XGBoost, a machine learning library. They found a substantial performance advantage for XGBoost, notably a reduced feature set containing fewer than one-third of the original variables, can achieve a nearly identical predictive performance.

Machine learning applications have been further extended to optimization in outfield defensive positioning. A study analyzing over fifty-one thousand batted balls from the 2023 MLB season evaluates how outfield positioning influences defensive outcomes. Results show that for ever one foot increase in straight line distance between a batted ball and the fielder, the odds of a hit increase by about five percent. Hwang (2025) also found that higher exit velocity and steeper launch angles are associated with a lower probability of a base hit. However, their findings also suggest that MLB outfielder positions are already near-optimal in most cases. The minimal difference in machine-learning metrics and actual defensive positions show a convergence towards a positional equilibrium. Teams independently arrive at similar spatial strategies.

A recent study using large language models to generate baseball spray charts. Using collegiate play-by-play text, they constructed approximately ninety-five percent accurate spray charts when compared to video tracking. The study finds that aggregated spatial patterns are highly stable despite noisy inputs (Michielssen 2024). These results introduce the idea that hitter outcomes are not randomly distributed across the field, but rather structured in spatial density patterns. While spray charts are effective in summarizing batted ball tendencies, they aggregate continuous spatial variation into discrete regions of the field, limiting their ability to provide meaningful results when it comes to if their exists actual statistically significant clustering. Therefore, there is motivation to apply clustering methods of spatial analysis to batted ball locations in order move beyond simple descriptive visualization.

A complementary line of research suggests a spatial structure in baseball data, when observations are grouped rather than analyzed in isolation. Marcou (2020) focuses on pitchers and the quality of contact they allow. He finds that while individual metrics such as exit velocity show substantial variability, clustering reveals that pitches tend to fall into stable groups. For instance, sinker-heavy pitchers exhibit ground ball rates near fifty-nine percent, whereas other pitcher profiles show roughly forty percent ground ball rates. Building on this idea, our project shifts the analysis from pitches to batters themselves. By applying spatial clustering methods, we aim to determine whether hitters exhibit similar latent profiles in terms of where they put the ball in play.

Oda and Hirotsu (2026) extend on Marcou's research as they develop a framework for classifying fielders in the Nippon Professional Baseball league using Gaussian model clustering. Using a dataset of one-hundred and fifteen hitting-related variables over three-hundred and twenty-seven players. They reduce multidimensional via principal component analysis, filtering for components that explain about seventy percent of the variance in performance categories related to plate discipline and win probability. This study finds that players grouped within the same cluster exhibit similar latent performance profiles. The authors multidimensional approach captures nuances that are overlooked by individual metrics. Our project moves the clustering work done by Oda from feature space to the physical space. The core argument that one-hundred and fifteen hitting indices can be condensed into distinct latent dimensions supports our aggregation. The spatial clustering we aim to perform reveals latent geographic structure in batted-ball locations.

Building on this shift from feature-space to physical-space, Cox (2010) provides a direct methodological precedent for spatial analysis of batted-ball locations in the MLB. Cox performs a spatial clustering of batted ball locations, through a hot spot analysis and Moran's I for significance. Cox examines manually collected data from the 2009 season at the Texas Rangers ballpark, pre-statcast. She goes on to perform density surface estimation and test for autocorrelation. This analysis lays the groundwork for our project. Her analysis was confined to a single ballpark and balls that left the field of play, however she notes that the framework could be extended to any batted-balls. Our project applies spatial clustering methods to all batted-balls in play across a stratified sample. This study lays the groundwork for methodology that we aim to extend to a different question, using recent statcast data. While Cox establishes that spatial clustering of batted-balls is feasible, a parallel study formalizes why space must be modeled continuously.

Another study that builds on the spatial structure in batting outcomes examines batting ability as a spatial Gaussian field with isotropic covariance. Cross and Sylvan (2015) empirically demonstrate that batting performance exhibits spatial dependence. They demonstrate through Monte Carlo simulations that traditional grid-based heat maps produce inaccurate estimate of batter ability. They found that a spatial analysis substantially outperforms discrete summaries. This study demonstrates the methodological value of treating space continuously, a vital component to our analysis. Extending this line of work, Comiskey (2017) develops variable-resolution heat maps that adapt to changing density across the strike zone. This analysis addressed the computational bottleneck that Bayesian spatial models face with batted-ball data. Combined, these works astablish batting outcomes requiring methods beyond discrete summarization. This study directly reinforces our analysis choice aims to utilize Kernal Density Estimations over fixed grids. Our point-pattern and clustering approach serve as a complementary methodology.

Healey (2017) develops a Bayesian model that uses kernel density estimation to construct a continuous mapping from batted-ball parameters including exit velocity and launch angle, extending this continuous space framework. Healey's use of nonparametric density estimation reinforces the need to treat batted-alls as a continuous spatial phenomenon. Our analysis recognizes that meaningful interpretation of our clusters requires accounting for underlying contact and other variables, outside of simply observed outcomes.

Recent work has begun applying formal point pattern methods directly to batted-ball locations. Kaji and Sylvan (2023) apply these patterns to model the inhomogeneous Poisson process underlying batted-ball distributions. They use statcast data and compare hitters by handedness. This study explores kernel density smoothing with optimal bandwidth selection, producing heat maps that visually distinguish the spatial patterns of right-handed and left-handed hitters. They only test three players, whereas our study expands this to fifteen players. Our study further builds directly on this framework but extends it to a different stratification of hitters, density based clustering, and significance testing to assess clustering patterns and if or how they deviate from spatial randomness.

Closely related, Young (2020) models the spatial distribution of batted-balls directly through spray-chart densities. They develop a Synthetic Estimate Average Matchup (SEAM) methodology which estimates the conditional distribution of balls-in-play for batter-pitcher matchups. This study incorporates player characteristics across similar players to model the structured probability distributions of different matchups. Young highlights the importance of density estimation techniques in capturing these patterns. Our analysis shifts this study from estimating conditional spray distributions to identifying spatial clustering.

Although these density studies establish that batted-ball locations are spatially structured, few test how observed clusters deviate from spatial randomness. Hochstedler (2016) explores the application of spatiotemporal machine learning techniques in professional sports. Methods such as Voronoi tessellations and neural networks demonstrate how spatial data can inform decision-making in sports. His work highlights the importance of spatial structure in player and ball movement, however largely assumes the existence of meaningful spatial patterns. Building on this, our project seeks to formally test whether these spatial structures exist in batted-ball distributions, and further test the significance of these spatial structures.

Finally, an earlier line of work focuses on the visualization of spatial baseball data, and how batted-ball data can advance scouting reports. Superak (2011) represents batted-ball data through density plots and contour maps to identify patterns in hitting behavior. One route in which we aim to take this project is to examine how we can apply our spatial analysis into real-world decision-making such as optimizing defensive positioning or tailoring strategic alignments to hitter-specific tendencies.

Data and Methodology

Unit of Analysis and Data Preparation

This project's unit of analysis is individual fair-territory batted balls, including home runs. Each observation represents a ball hit by a player in fair territory during the 2023 MLB season and is treated as spatial point data. Statcast's hc_x and hc_y variables are image-based plotting coordinates and do not provide spatially interpretable field-distance coordinates. These variables were transformed into spray_x and spray_y to set home plate as the origin, extend the positive y-axis toward center field, and position the x-axis to capture horizontal location (negative left, positive right).

Sampling

To construct the player sample, MLB hitters with at least 500 plate appearances were identified, producing an initial pool of 136 players. This threshold was put in place to ensure each player had sufficient batted-ball events for spatial analysis. Then, players were classified into a directional archetype using their pull, center, and opposite-field percentages. If one of these directional percentages exceeded 40%, the player was classified as a pull, center, or opposite-field hitter based on their principal directional tendency. If none exceeded 40%, then the player was classified as balanced. This yielded 75 pull, 51 balanced, 9 center, and 1 opposite-field hitter. A stratified sampling approach was implemented to preserve the distribution of directional archetypes in the selection of 15 players: 8 pull, 5 balanced, 1 center, and 1 opposite-field hitter were randomly selected. Center and opposite-field hitters were rounded up to the nearest integer; balanced and pull hitters were rounded down to the nearest integer to ensure representation of all directional archetypes and sample size.

Spatial and Directional Analysis

This project utilizes statistical tests and spatial visualizations to analyze the data. First, a player's batted balls were plotted using the transformed spray_x and spray_y coordinates on the approximated baseball field. Each batted ball was then color classified by the result of the play (single, double, triple, home run, out, or error). Next, the field was divided into 9 subregions based on depth and horizontal alignment. The observations in each subregion were counted to compare and help determine in which areas of the field a hitter's batted balls tend to concentrate. A χ^2 test was then conducted to evaluate if observed batted-ball distributions significantly deviated from a spatially uniform distribution of balls in each of the 9 zones. Next, spatial autocorrelation was evaluated using Global Moran's I and Local Indicators of Spatial Autocorrelation (LISA). Global Moran's I was tested to determine if subregions with similar batted-ball counts were spatially clustered. LISA was used to determine where significant spatial autocorrelation exists locally and to produce LISA visualizations. Kernel density estimation was used to create continuous density surfaces highlighting areas where each player's batted balls tend to cluster.

The Ripley's L function was used to analyze batted-ball clustering by using the distance between observations to determine whether points are more clustered or dispersed compared to complete spatial randomness (CSR). Each player's Ripley's L curve was compared to a Monte Carlo envelope constructed by simulating random point patterns within the study area. If the observed Ripley's L curve was above the envelope at a given distance, this indicated that points exhibited more spatial clustering than expected by CSR. If the pattern fell below the envelope, then the observed points were more dispersed at that given distance. A normalized spatial entropy index was calculated using the hitters' 9-zone batted ball proportions to summarize overall concentration. Shannon entropy was selected as it provides a single scalar measure of distributional concentration that is directly comparable across players. A lower entropy signifies that a hitter's batted balls are concentrated in fewer zones, and a higher entropy indicates a more even distribution across the study area. Normalization is accomplished by dividing Shannon entropy by the maximum possible entropy across the nine zones.

Software

All the data analysis and visualizations were conducted in R Studio using R. Statcast batted-ball data were accessed through the baseballr package (Petti & Gilani, 2024). FanGraphs directional hitting data were used to classify hitters by batted-ball tendency. Batter archetype classification and stratified sampling were completed using FanGraphs data in Excel before the selected sample was analyzed in R (Weinberg, 2015).

Results and Discussion

Conclusion

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Appendix